INCH-POUND

MIL-DTL-81940C
17 October 2019
SUPERSEDING
MIL-V-81940B
30 August 1993

DETAIL SPECIFICATION

VALVE, SAMPLING AND BLEED, HYDRAULIC, TYPE II SYSTEMS

This specification is approved for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers hydraulic sampling and bleed valves suitable for use in aircraft and missile high-pressure hydraulic systems.

1.2 Classification. Hydraulic sampling and bleed valves are of class 3000 and are to be operated with type II hydraulic fluid as defined in SAE AS5440.

2. APPLICABLE DOCUMENTS

2.1 General. The documents listed in this section are specified in sections 3 and 4 of this specification. This document does not include documents cited in other sections of this specification or recommended for additional information or as examples. While every effort has been made to ensure the completeness of this list, document users are cautioned that they must meet all specified requirements of documents cited in sections 3 and 4 of this specification, whether or not they are listed.

2.2.1 Specifications, standards and handbooks. The following specifications, standards and handbooks form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those listed in the solicitation or contract.

Comments, suggestions, or questions on this document should be addressed to (Naval Air Warfare Center, Aircraft Division Lakehurst, Code 4.1.2.2, Mail Stop 120-3, Route 547, Joint Base MDL, NJ 08733-5100) or emailed to michael.sikora@navy.mil. Since contact information can change, you may want to verify the currency of this address information using the ASSIST Online database at https://assist.dla.mil.

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DEPARTMENT OF DEFENSE SPECIFICATIONS

MIL-PRF-5606	-	Hydraulic Fluid, Petroleum Base; Aircraft, Missile and Ordnance
MIL-PRF-6083	-	Hydraulic Fluid, Petroleum Base, For Preservation and Operation
MIL-PRF-81836	-	Filter and Disposable Element, Fluid Pressure, Hydraulic, 3 Micro Absolute, General Specification for
MIL-DTL-81940/1	-	Valve, Sampling and Bleed, Hydraulic, Type II Systems
MIL-DTL-81940/2	-	Valve, Sampling and Bleed, Hydraulic, Type II Systems NPTF Fitting
MIL-DTL-81940/3	-	Valve, Sampling and Bleed, Hydraulic, Type II Systems 37° Flared Fitting
MIL-DTL-81940/4	-	Valve, Sampling and Bleed, Hydraulic, 3000 PSI Non-Aircraft Applications
MIL-PRF-83282	-	Hydraulic Fluid, Fire Resistant, Synthetic Hydrocarbon Base, Aircraft, Metric, NATO Code Number H-537
MIL-PRF-87257	-	Hydraulic Fluid, Fire Resistant, Low Temperature, Synthetic Hydrocarbon Base, Aircraft and Missile

DEPARTMENT OF DEFENSE STANDARDS

MIL-STD-130	-	Identification Marking of U.S Military Property
MS21344	-	Fittings, Installation of Flared Tube, Straight Threaded Connectors, Design Standard For

DEPARTMENT OF DEFENSE HANDBOOK

MIL-HDBK-505	-	Definitions of Item Levels, Item Exchangeability, Models and Related Terms

(Copies of these documents are available online at https://quicksearch.dla.mil.)

2.2.2 Other Government documents. The following other Government documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

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CODE OF FEDERAL REGULATIONS (CFR)

40 CFR 82 - Protection of Stratospheric Ozone

(Copies of this document are available online at http://www.ecfr.gov.)

2.3 Non-Government publications. The following documents form a part of this document to the extent specified herein. Unless otherwise specified, the issues of these documents are those cited in the solicitation or contract.

AMERICAN SOCIETY OF MECHANICAL ENGINEERS (ASME)

ASME Y14.100 Appendix D - Engineering Drawing Practices; Numbering, Coding and Identification

(Copies of this document are available from https://www.asme.org.)

AMERICAN SOCIETY FOR QUALITY (ASQ)

ASQ-Z1.4 - Sampling Procedures and Tables for Inspection by Attributes

(Copies of this document are available from www.asq.org.)

INTERNATIONAL ORGANIZATION FOR STANDARIZATION (ISO)

ISO 11171 - Hydraulic Fluid Power – Calibration of Automatic Particle Counters for Liquids

ISO 12103-1 - Road Vehicles – Test Contaminants Filter Evaluations Part 1: Arizona Test Dust

(Copies of this document are available online at www.iso.org.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE) INTERNATIONAL

SAE AS598 - Aerospace Microscopic Sizing and Counting of Particulate Contamination for Fluid Power Systems

SAE AS4059 - Aerospace Fluid Power - Contamination Classification for Hydraulic Fluids

SAE AS5440 - Hydraulic Systems, Military Aircraft, Design and Installation Requirements for

SAE AS8775 - Hydraulic Systems Components, Aircraft and Missile, General Specification for

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(Copies of these SAE documents are available from www.sae.org.)

2.4 Order or precedence. Unless otherwise noted herein or in the contract, in the event of a conflict between the text of this document and the references cited herein, the text of this document take precedence. Nothing in this document, however, supersedes applicable laws and regulations, unless a specific exemption has been obtained.

3. REQUIREMENTS

3.1 Specification sheets. The individual item requirements shall be as specified herein and in accordance with the applicable specification sheet. In the event of any conflict between the requirements of this specification and the specification sheet, the latter shall govern.

3.2 Qualification. Hose assemblies furnished under this specification shall be products that are authorized by the qualifying activity for listing on the applicable Qualified Products List before contract award (see 4.2 and 6.3).

3.3 Materials.

3.3.1 Selection. Materials used shall be internally and externally protected against corrosion. Material selection shall be in accordance with SAE AS8775, Type II components.

3.3.2 Compatibility. The valve shall be constructed of materials that will not adversely affect or be affected by hydraulic fluid conforming to MIL-PRF-5606, MIL-PRF-83282, MIL-PRF-87257 and the test fluids specified herein. Valves in non-aircraft applications shall be constructed of materials that will not adversely affect or be affected by hydraulic fluid conforming to MIL-PRF-6083.

3.3.3 Fungi nutrients. Materials that are not nutrients for fungi shall be used wherever possible. Where fungus-nutrient materials must be used, they shall be treated with a fungicidal agent.

3.3.4 Prohibited materials. Class I and II Ozone Depleting Substances (ODS) as defined in 40 CFR 82 shall not be used in MIL-DTL-81940 or any referenced procedures.

3.4 Design and construction.

3.4.1 Sampling and bleed valves. Sampling and bleed valves, for hydraulic systems conforming to SAE AS5440 for aircraft shall be designed and constructed as specified herein and the applicable specification sheet.

3.4.2 External loads. The valve shall be designed to withstand all the external loads. The valve shall be of such strength and rigidity as to withstand the wrench loads required for making tube connections. The installation torque and tube connection shall be in accordance with MS21344.

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3.4.3 Valve internal configuration. The internal configuration of the valve shall be inspected and shall be such that the minimum restriction in the flow path shall be 500 microns.

3.5 Performance. The valve shall meet the performance requirements as specified in 3.5.1 through 3.5.12.

3.5.1 Sample efficiency (normal systems). The valve shall be designed to operate with hydraulic fluids with a contamination level as specified in 4.5.1.1. For each size range, the particle count from the sampling valve shall be equal to the particle count from the open port within ± 20 percent.

3.5.2 Sampling efficiency (dirty systems). The valve shall be designed to operate with hydraulic fluids having a contamination level higher than normal (class 12 and above contamination level per SAE AS4059) hydraulic fluid systems and shall be able to meet the test requirements as specified in 4.5.1.2.

3.5.3 Proof pressure. The valve shall be designed to withstand a proof pressure of 6000 psi and shall not leak, malfunction, or permanently deform when tested in accordance with proof pressure test as specified in 4.5.2.

3.5.4 Leakage (uncapped). The valve shall be designed to operate at 3000 psi with no cap installed in the outlet port and shall meet the leakage test requirements as specified in 4.5.3.

3.5.5 Leakage (capped). The valve shall be designed to operate at 3000 psi with a cap installed in the outlet port and shall be able to meet the leakage test requirements when tested in accordance with 4.5.4.

3.5.6 Valve operation. The valve shall be designed to operate at 3000 psi continuously for 2000 hours at 275 ºF fluid and ambient temperature without malfunction or leakage and shall meet the test requirements of 4.5.5.

3.5.7 Low temperature operation. The valve shall be designed to operate at -70 ºF (maximum) hydraulic fluid and ambient temperature and shall be able to pass the test requirements as specified in 4.5.6.1.

3.5.8 High temperature operation. The valve shall be designed to operate at 275 ºF (maximum) hydraulic fluid and ambient temperature without malfunction or leakage. The valve shall meet the test requirements as specified in 4.5.6.2.

3.5.9 Impulse. The valve shall be able to operate at the peak pressure of 150 percent of operating pressure with initial rate of pressure rise to 200,000-300,000 psi/sec and shall meet the test requirements for the impulse test (see 4.5.7).

3.5.10 Vibration. The valve shall be designed to meet the vibration test requirements as specified in 4.5.8.

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3.5.11 Burst pressure. The valve shall be designed to meet the burst pressure of 12,000 psi and shall show no leakage when held at that pressure for 2 minutes (see 4.5.9).

3.5.12 Cleaning. Prior to preservation or packaging, the valve shall be cleaned ultrasonically (or equivalent method) until free from all metal chips, grit, dirt, and other foreign matter. The cleaning method utilized shall not have a deleterious effect on any material (e.g, metals, plastics, elastomers, etc.). Care shall be taken after cleaning to ensure that the valve is not contaminated prior to or during preservation or packaging.

3.6 Identification of product.

3.6.1 Markings. Each valve shall be clearly and permanently identified by steel stamping or by a permanently attached nameplate conforming to MIL-STD-130. The following information shall be provided:

VALVE, SAMPLING AND BLEED
M81940/ (Insert slash number of specification sheet and designation for fitting end)
Manufacturer part number
Manufacturer stock number or serial number
Manufacturer name
Design activity CAGE code/part number

3.7 Part numbering of interchangeable parts. All parts having the same design activity CAGE (Commercial and Government Entity) code and part number shall be interchangeable. MIL-HDBK-505 gives guidance on the interchangeability of parts. The item identification and part number requirements of ASME Y14.100, Appendix D shall govern the design activity part numbers and changes thereto. The manufacturer's part number and drawing number shall be the same.

3.8 Workmanship. The workmanship and finish shall be of sufficiently high grade to ensure satisfactory operation, reliability, and durability consistent with the application and storage life requirements of the valve.

4. VERIFICATION

4.1 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. Qualification inspection (4.2).

b. Conformance inspection (4.3).

4.2 Qualification inspection. The qualification inspection shall consist of all the examinations and tests specified in table I (listed sequence mandatory).

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4.2.1 Qualification samples. The qualification sample shall consist of two specimens of each valve. The specimens shall be assembled of parts that conform to manufacturer's drawings/design activity drawings. Samples shall be forwarded to a test facility (see 6.3). Samples shall be clearly identified by securely attached durable tags with the following information:

> Sample submitted by (name) (date) for qualification inspection in accordance with the requirements of MIL-DTL-81940 and 6.3.

4.2.2 Retention of qualification. The retention of qualification shall consist of verification every two years to determine compliance of the listed items with the requirements of this specification. Verification shall be by manufacturer's certification unless otherwise specified by the activity responsible for the QPL and shall be included in the notice of qualification letter.

4.3 Conformance inspection. Conformance inspection shall consist of the following:

a. Individual inspection (4.3.1).
b. Sampling tests (4.3.3).

4.3.1 Individual inspection. The individual inspection specified in table II shall be conducted on each valve in the order listed. Any valve containing defects will be rejected.

TABLE I. Qualification inspection.

Test in sequence	Test description	Requirements paragraph no.	Test requirements paragraph no.
1	Examination of product	3.8	4.3.2
2	Valve cleanliness	3.5.12	4.6
3	Sampling efficiency (normal system)	3.5.1	4.5.1.1
4	Sampling efficiency (dirty system)	3.5.2	4.5.1.2
5	Proof pressure	3.5.3	4.5.2
6	Leakage (uncapped)	3.5.4	4.5.3
7	Leakage (capped)	3.5.5	4.5.4
8	Valve operation	3.5.6	4.5.5
9	Low temperature operation	3.5.7	4.5.6.1
10	High temperature operation	3.5.8	4.5.6.2
11	Impulse	3.5.9	4.5.7
12	Vibration	3.5.10	4.5.8
13	Burst pressure	3.5.11	4.5.9

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4.3.2 Examination of products. Each valve shall be carefully examined to determine conformance with the requirements of this specification and specification sheets with regard to design, construction, workmanship, and product conformance to applicable drawings and for any visible defects.

4.3.3 Sampling tests. A sample shall be selected from each inspection lot (see 4.3.4) in accordance with ASQ-Z1.4 using Special Inspection Level S-1 with no rejects allowed. The sample valve shall be unpacked and the sampling tests, specified in table II, shall be conducted on each valve in the order listed.

TABLE II. Conformance inspection.

Test sequence	Test description	Requirements paragraph no.	Test requirements paragraph no.
	INDIVIDUAL INSPECTION		
1	Examination of product	3.8	4.3.2
2	Proof pressure at room temperature	3.5.3	4.5.2
	SAMPLING TESTS		
1	Valve cleanliness	3.5.12 & 4.4.1.1	4.6
2	Leakage	3.5.4	4.5.3

4.3.4 Inspection lot. For purposes of conformance inspection, a lot shall be defined as all units of product manufactured under essentially the same conditions and offered for acceptance at the same time.

4.4 Test conditions.

4.4.1 Test fluid. Unless otherwise specified, the hydraulic fluid used for all tests shall conform to MIL-PRF-83282 for the -40 ºF to +275 ºF temperature range or MIL-PRF-5606 for the -65 ºF to +275 ºF temperature range (with no free water). For conformance tests of valves in non-aircraft applications, hydraulic fluid conforming to MIL-PRF-6083 shall be used.

4.4.1.1 Test fluid cleanliness. For conformance tests and the cleanliness qualification test (see 4.6), the test fluid shall be pre-cleaned to a cleanliness level equal to class 1 of SAE AS4059.

4.4.1.2 Test fluid filtration. The test fluid shall be continuously filtered through a 3-micron absolute filter conforming to MIL-PRF-81836 or a filter with equivalent efficiency during testing.

4.4.1.3 Test fluid temperature. The actual temperature of the test fluid shall be 100 ºF ±10 ºF.

4.4.2 Test temperature. The tests shall be conducted at a room temperature of 70 ºF to

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90 ºF, measured within 12 inches of the test sample.

4.5 Test methods.

4.5.1 Valve characteristics.

4.5.1.1. Valve sampling efficiency (normal system). The clean-up filter on the figure 1 system shall be bypassed. ISO Medium Test Dust in accordance with ISO 12103-1 shall be added to the system until a contamination level of between class 6 and 8 of SAE AS4059 is obtained. The fluid sample for the above shall be drawn from the open port shown on figure 1. Flow shall be adjusted to 5 gpm with 400 psi maintained on the valve. The valve shall be uncapped, actuated and 200 ml flushed through the valve sampling port. With the valve kept in an open position, 100 ml of fluid shall be collected in a pre-cleaned bottle. The bottle shall be removed from the fluid stream prior to closing the valve. The fluid shall be analyzed in accordance with SAE AS598 or with an automatic particle counter that has been calibrated using ISO 11171.

4.5.1.2 Valve sampling efficiency (dirty system). The test specified in 4.5.1.1 shall be repeated except that ISO Medium Test Dust in accordance with ISO 12103-1 shall be added to the system until a contamination level of class 12 or greater is obtained. Flow through the sampling valve shall be continuous with no blockage. Five 100 ml samples shall be collected. The sampling valve flow rate shall be as specified in 4.5.5. Circulate the system on figure 1 for 12 hours with the clean-up filter in bypass mode and repeat this test. Any blockage or discontinuous flow during the test shall be cause for rejection.

4.5.2 Proof pressure. The valve shall be filled with test fluid and maintained at 275 ºF for 72 hours for the qualification test only. A proof pressure of 6,000 psi shall then be applied at least twice at 275 ºF and held for 2 minutes at each application. The pressure shall be reduced to zero between pressure applications. The activity responsible for qualification shall have the prerogative to conduct additional tests to validate compliance of the valve with the longevity requirements specified in 3.5.6, and verify the manufacturer's analysis. For quality conformance testing, a proof pressure (6,000 psi) shall be applied at least twice at room temperature of 70 ºF and held for 2 minutes at each application. The pressure shall be reduced to zero between pressure applications. Any leakage or malfunction during the test shall constitute failure of the test and rejection of the valve.

4.5.3 Leakage (uncapped valve). The valve shall be uncapped and subjected to 3,000 psi. The valve shall be actuated and a 100 ml sample of fluid collected. The valve knob or button shall be released and allowed to close by itself. The 3,000 psi pressure shall be maintained for 2 hours. There shall be no leakage from the valve. The test shall be repeated at 5 psi. The valve shall automatically and immediately close in each case and no leakage shall occur. Any leakage in the valve during the test shall constitute failure of the test and rejection of the valve.

4.5.4 Leakage (capped valve). The valve with cap installed shall be subjected to 3,000 psi. The valve shall be held in an open position for 2 hours. A mechanical means can be used to hold the valve open. There shall be no leakage from the valve, or from behind the push button or

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knob. Any leakage in the valve during the test shall constitute failure of the test and rejection of the valve.

4.5.5 Valve operation. The valve tested shall be installed in the test set up as shown on figure 1. Flow shall be adjusted to 5 gpm with 3,000 psi maintained on the valve. The valve shall be uncapped and actuated to a full open position. The force or torque, as applicable, required to actuate the valve fully shall be measured and shall not exceed the value specified on the specification sheet. The valve knob shall be opened fully and a 100 ml sample of fluid collected. The valve shall be allowed to close. The flow rate shall be between 100 and 1500 ml per minute. The velocity of fluid exiting from the discharge port shall be 30 feet per second at full open position. The valve shall be actuated 100 times with 100 ml of fluid collected each time. The valve shall be tested for leakage in accordance with 4.5.3. The above test shall be repeated with 50 psi maintained on the valve. Any malfunction or fluid flow less than 30 ft. per second shall constitute failure of the test and rejection of the valve.

4.5.6 Extreme temperature functioning.

4.5.6.1 Low temperature functioning. The valve shall be uncapped and subjected to -65 ºF to -70 ºF for 24 hours. During this period, the valve shall be filled with MIL-PRF-5606 hydraulic fluid. With fluid and ambient temperature maintained at -65 ºF to -70 ºF the tests specified in 4.5.3 and 4.5.4 shall be conducted. There shall be no leakage from the valve. Any fluid leakage during the test shall constitute failure of the test.

4.5.6.2 High temperature functioning. The valve shall be subjected to a temperature of 275 ºF ±5 ºF for 24 hours. During this period, the valve shall be filled with test fluid. With fluid temperature at 275 ºF ±5 ºF, the tests specified in 4.5.3 and 4.5.4 shall be conducted. There shall be no leakage from the valve. Any fluid leakage during the test shall constitute failure of the test.

4.5.7 Impulse. The valve shall be subjected to 100,000 impulse cycles, 25,000 at a fluid temperature of 275 ºF and 75,000 at a fluid temperature of 225 ºF. Each impulse cycle shall consist of a pressure rise from zero to 3,000 psi and back to zero. During each pressure increase, a peak surge pressure of 1.43 to 1.57 times the working pressure, as shown by an oscillograph, shall be obtained. Cycling shall be performed at a rate of 300 cycles per minute (cpm) maximum. There shall be no evidence of external leakage or structural failure during the performance of this test. The impulse pattern shall be as shown on figure 2. Any fluid leakage during the test shall constitute failure of the test.

4.5.8 Vibration. A sinusoidal vibration test shall be conducted on the valve at room temperature on each of the valve's three mutually perpendicular axes. One of these axes shall be parallel to the valve poppet axis. The valve shall be mounted on a resonance-free fixture for the test. The vibration input accelerometer shall be mounted on the fixture close to the mounting point of the valve. A second accelerometer shall be mounted directly on the valve for monitoring resonant frequencies. The rate of change of frequency shall be logarithmic (see figure 3). Any leakage during the test shall constitute failure of the test.

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Vibration tolerances	
Amplitude	± 10%
Frequency	± 2%
Acceleration	± 10%

The following tests shall be conducted:

a. Resonance search. Resonant frequencies of the valve assembly shall be determined by varying the frequency of applied vibration slowly through the range specified in table III, part 1, at reduced levels but with sufficient amplitude to excite them. This shall be repeated for each of the three axes.

b. Resonance dwell. The valve shall be vibrated along each axis at the four most severe resonant frequencies (determined in the resonance search) for 30 minutes at each resonance. The vibration levels and test times shall be in accordance with table III, part 1.

c. Cycling test. The valve shall be vibrated along each axis in accordance with the test levels, frequency ranges and times from table III, parts 1 and 2. The frequency of applied vibration shall be swept over the frequency range of 5 to 2000 Hz and back to 5 Hz logarithmically. The time to sweep from 5 to 2000 Hz and back to 5 Hz shall be approximately 20 minutes. If a change in the resonant frequency occurs during the test, the time of occurrence shall be recorded and the frequency shall be adjusted immediately to maintain peak resonance condition. The final resonant frequency shall be recorded.

d. Leakage test. Following the vibration test the valve shall be subjected to the leakage tests specified in 4.5.3 and 4.5.4. There shall be no leakage from the valve.

4.5.9 Burst pressure. Pressure shall be applied with a pump at a maximum rate of increase of 25,000 psi per minute until the specified burst pressure of 12,000 psi is reached. The valve shall show no leakage in the form of drops or rupture of internal or external parts at this pressure when held for 2 minutes. The pressure may be increased above that specified during qualification tests in order to secure data on actual burst pressure. The test shall be conducted at 275 ºF after a 5-hour soak at 275 ºF, and the valve filled with test fluid.

4.6 Valve cleanliness. The test fluid used in determining sampling valve characteristics (see figure 1) shall be pre-cleaned in accordance with 4.4.1.1. The valve shall be removed from its packaging and immediately installed on the figure 1 system. Flow shall be adjusted to 5 gpm through the system with 400 psi maintained on the valve. The valve shall be uncapped, actuated and 200 ml flushed through the valve sampling port. With the valve kept in an open position, 100 ml of fluid shall be collected in a pre-cleaned bottle. The bottle shall be removed from the fluid stream prior to closing the valve. The fluid shall be analyzed in accordance with SAE AS598 or with an automatic particle counter that has been calibrated using ISO 11171. The amount of contaminant shall not exceed class 1 of SAE AS4059. Fluid contamination above class 1 of SAE AS4059 shall constitute failure of the test.

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5. PACKAGING

5.1 Packaging. For acquisition purposes, the packaging requirements shall be as specified in the contract or order (see 6.2). When packaging of materiel is to be performed by DoD or in-house contractor personnel, these personnel need to contact the responsible packaging activity to ascertain packaging requirements. Packaging requirements are maintained by the Inventory Control Point's packaging activities within the Military Service or Defense Agency, or within the military service's system commands. Packaging data retrieval is available from the managing Military Department's or Defense Agency's automated packaging files, CD-ROM products, or by contacting the responsible packaging activity.

6. NOTES

(This section contains information of a general or explanatory nature which may be helpful, but is not mandatory.)

6.1 Intended use. The valve covered by this specification is intended for use in aircraft and missile hydraulic systems covered by SAE AS5440 and operating with hydraulic fluid conforming to MIL-PRF-5606 or MIL-PRF-83282 at a nominal operating pressure of 3,000 psi. For low pressure systems ground support application, see MIL-DTL-81940/2 and MIL-DTL-81940/3 and for non-aircraft application, see MIL-DTL-81940/4.

6.2 Acquisition requirements. Acquisition documents should specify the following:

a. Title, number and date of this specification.

b. Packaging (see 5.1).

6.3 Qualification. With respect to products requiring qualification, awards will be made only for products which are, at the time of award or contract, qualified for inclusion in Qualified Products List QPL No. 81940 whether or not such products have actually been so listed by that date. The attention of the contractors is called to these requirements, and manufacturers are urged to arrange to have the products that they propose to offer to the Federal Government tested for qualification in order that they may be eligible to be awarded contracts or orders for the products covered by this specification. Information pertaining to qualification of products may be obtained from Commander, Naval Air Warfare Center Aircraft Division, Attn: Code 4.3.5, Building 3197, Suite D, 22347 Cedar Point Road, Unit 6, Patuxent River, MD 20670-1161. An online listing of products qualified to this specification may be found in the Qualified Products Database (QPD) at https://assist.dla.mil.

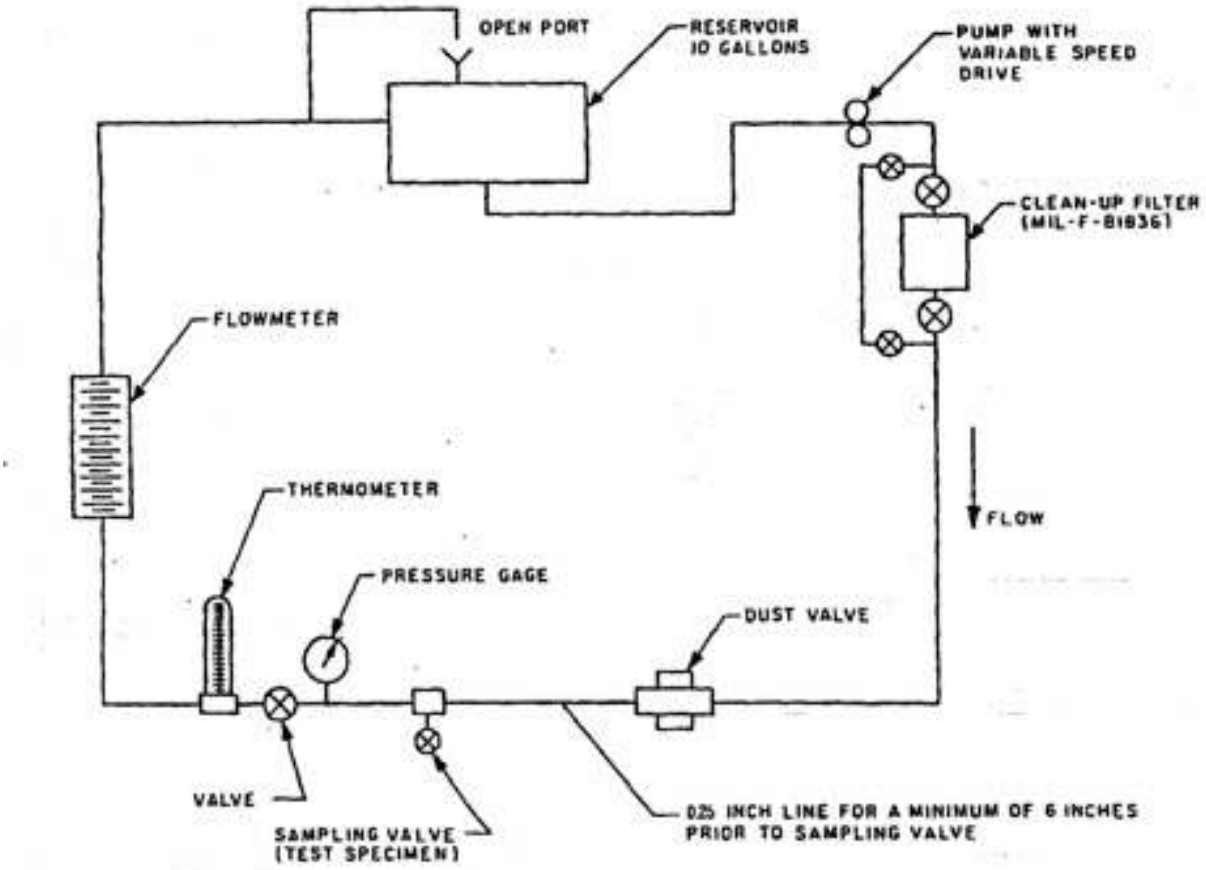
6.4 Subject term (key word) listing.

Aircraft, Control
Flow, Fluid
Flow, High pressure
Ground Support Equipment
Missiles, Weapons

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6.5 Changes from previous issue. The margins of this specification are marked with vertical lines to indicate where changes from the previous issue were made. This was done as a convenience only and the Government assumes no liability whatsoever for any inaccuracies in these notations. Bidders and contractors are cautioned to evaluate the requirements of this document based on the entire content irrespective of the marginal notations and relationship to the previous issue.

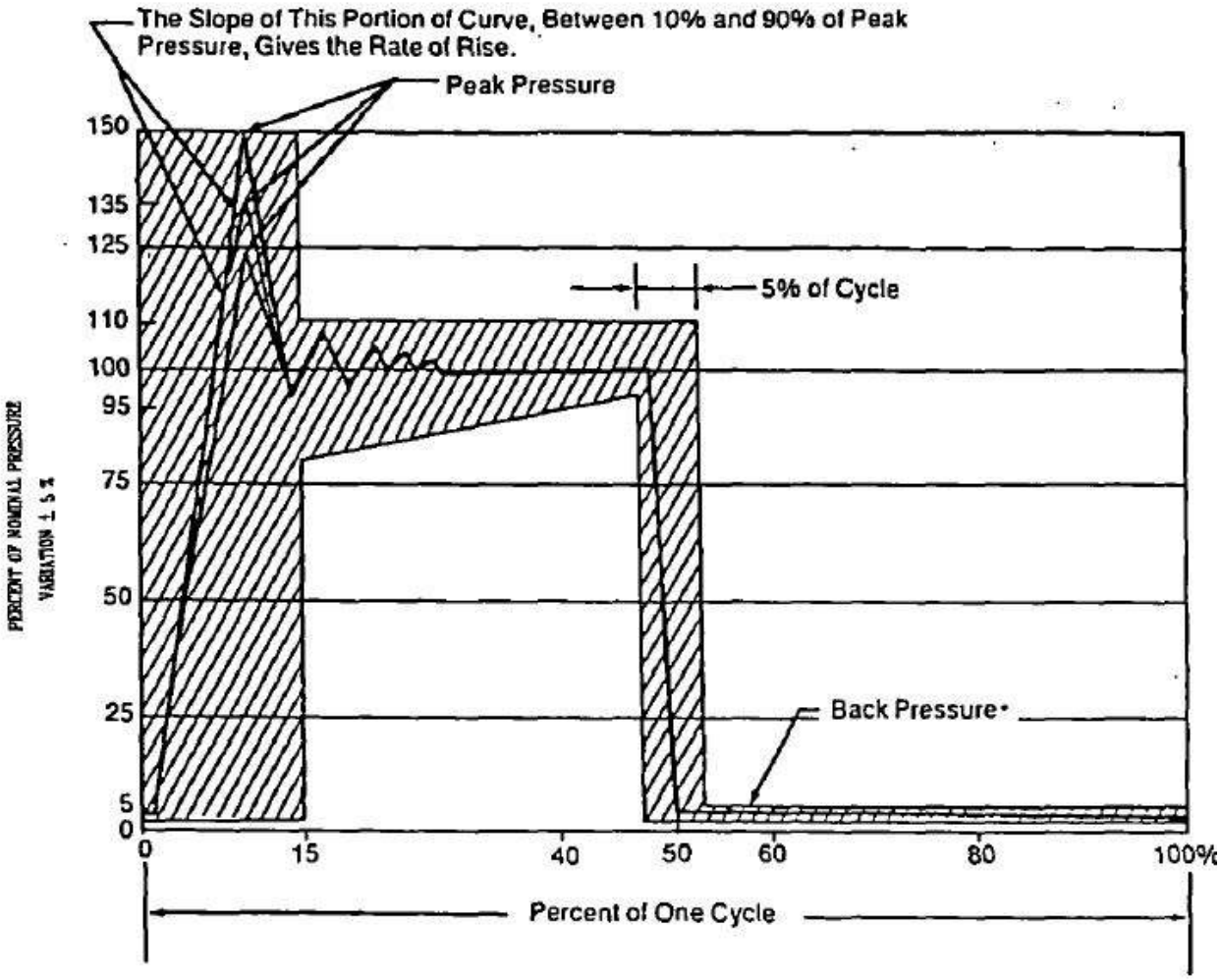
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NOTE: Total system volume shall not exceed 15 gallons.

FIGURE 1. Typical setup for determining sample valve characteristics.

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Note: The curve shown above is the approximate pressure-time cycle determined to be of proper severity for impulse testing. Although it is mandatory only that the pressure peak rises to 150% of the operating pressure at some point prior to leveling off at rated pressure, it is considered highly desirable that the pressure-time curve be confined to the shaded area indicated. Initial rate of pressure rise shall be 200,000 - 300,000 psi per second. The back pressure shall be 50 ±25 psi.

FIGURE 2. Impulse curve.

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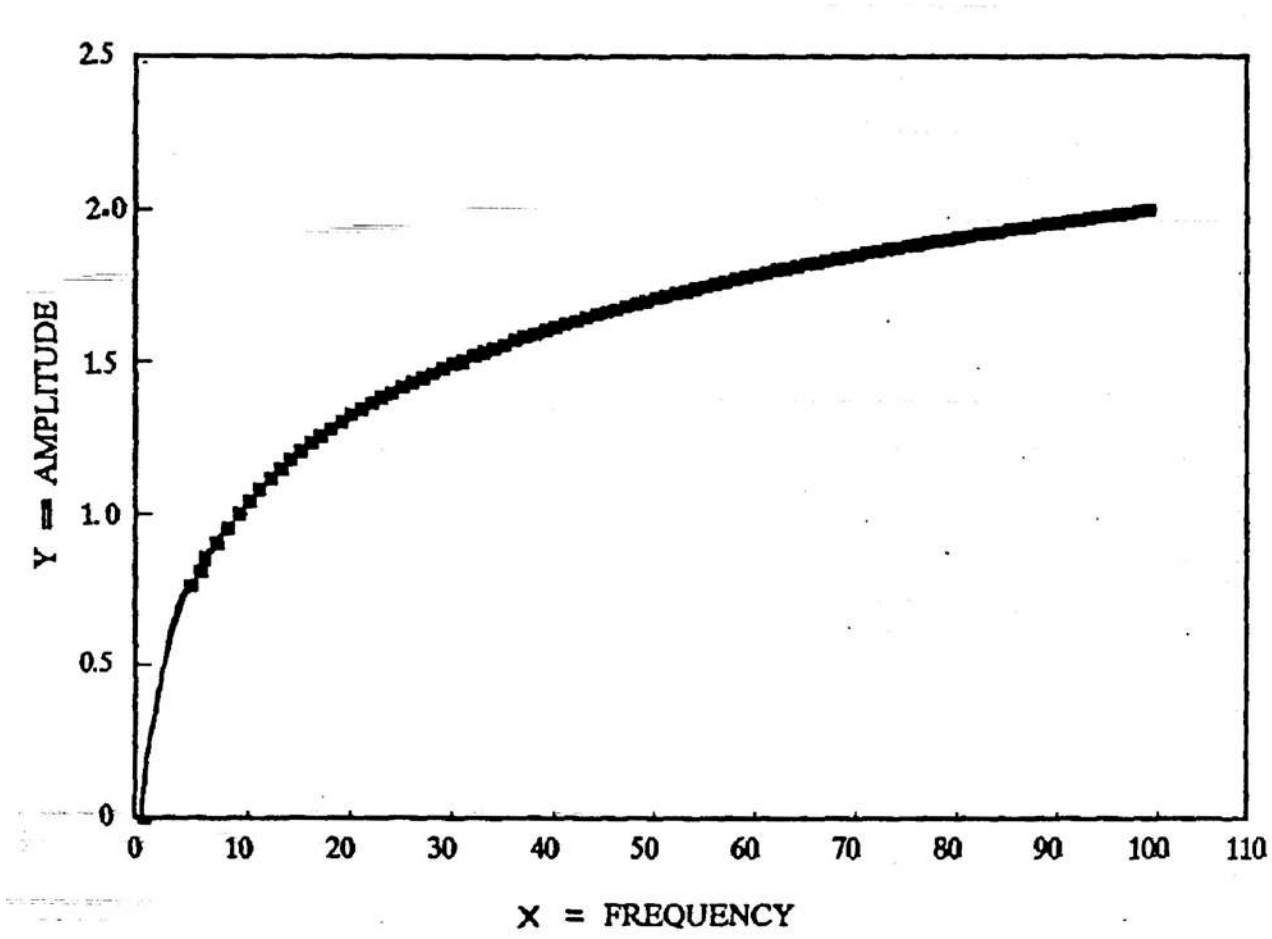


FIGURE 3. Logarithmic rate of change of frequency.

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TABLE III. Test procedure and time schedule.

Test	Vibration level					
Part 1	0.1 inch double amplitude 1.0 g 0.036 inch double amplitude 10.0 g				Frequency band (Hz) 5 to 14 14 to 23 23 to 74 74 to 2000	
Part 2	0.2 inch double amplitude 1.0 g 0.06 inch double amplitude 20.0 g				Frequency band (Hz) 5 to 10 10 to 18 18 to 81 81 to 2000	
Vibration time schedule - (time per one axis)						
	Number of resonances	0	1	2	3	4
Part 1	Total vibration time at each resonance	0	½ hr	1 hr	1-1/2 hr	2 hr
	Cycling time	2 hr	1-1/2 hr	1 hr	½ hr	0
Part 2	30 minutes cycling per axis no resonance dwell					

Custodians:
Army - AV
Navy - AS
Air Force - 71

Preparing activity:
Navy - AS

(Project 4820-2019-004)

Review activities:
DLA - CC
Navy - SH

NOTE: The activities listed above were interested in this document as of the date of this document. Since organizations and responsibilities can change, you should verify the currency of the information above using the ASSIST online database at https://assist.dla.mil.

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